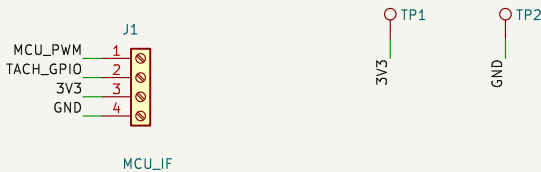
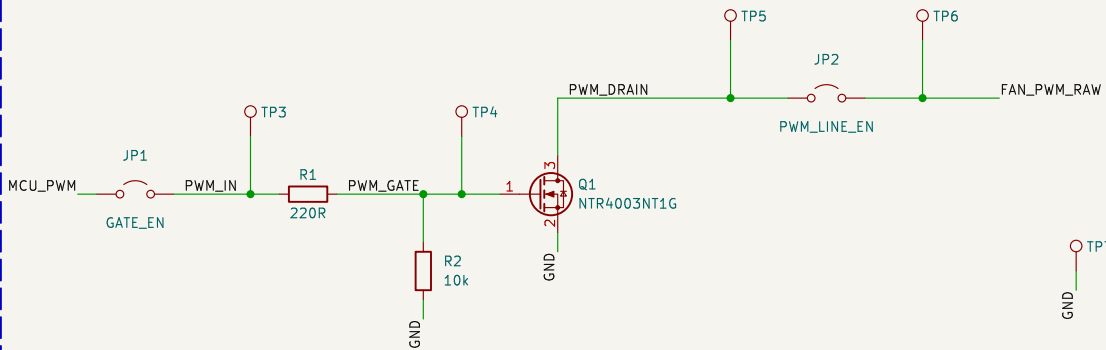


1. MCU / DragonBench interface



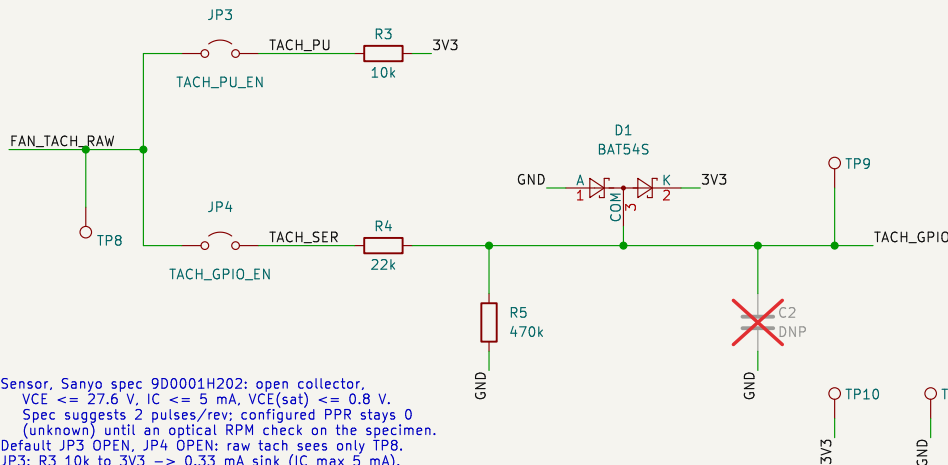
Controller-agnostic link. Board wiring is a LOCAL overlay.
N8RB bench overlay (provisional, NOT product GPIOs):
J1.1 MCU_PWM <- GPIO6
J1.2 TACH_GPIO -> GPIO7
J1.3 3V3 <- board 3V3 (never 5V)
J1.4 GND <- board GND
sdkconfig.fan-characterization.local:
CONFIG_DB_FAN_PWM_GATE_GPIO=6
CONFIG_DB_FAN_GATE_SINK_LEVEL=1
CONFIG_DB_FAN_TACH_GPIO=7
TinyS3[D]: same J1 functions, own overlay.
Breakout is passive: no protection assumed.

2. PWM open-drain driver



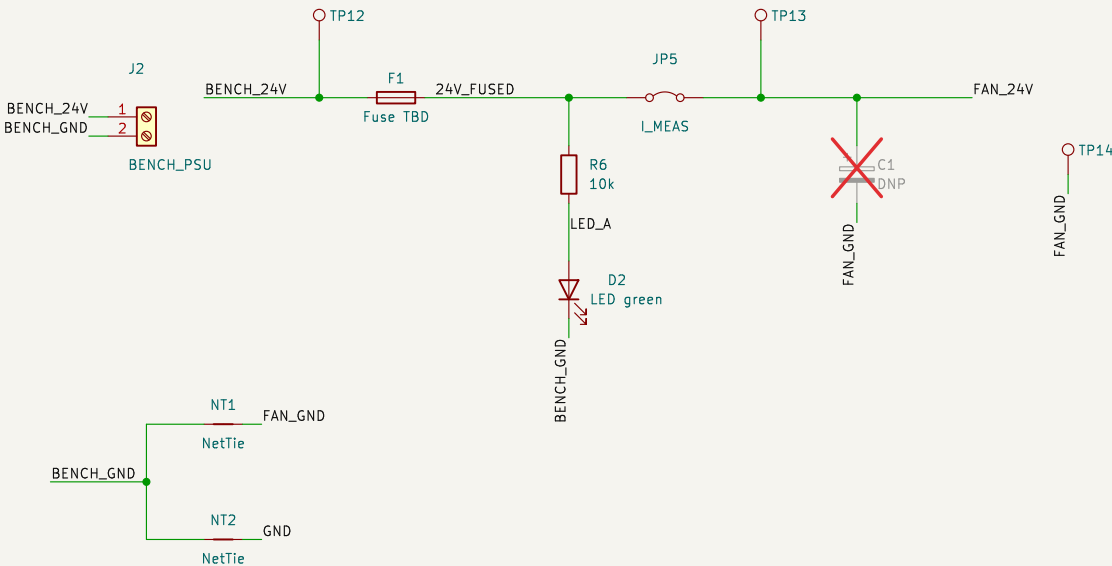
Q1 non-inverting: gate HIGH = sink => CONFIG_DB_FAN_GATE_SINK_LEVEL=1. sink_duty_pct = Q1 on-time.
Fan control input, 9GA0424P3J001 drawing Rev C: 25 kHz specified; open-collector/open-drain drive OK;
control open = same speed as 100 %; 0 % = stopped; open voltage <= 5.25 V; source <= 2 mA at 0 V; VIL <= 0.4 V.
Q1 NTR4003NT1G is a signal-interface FET here: normal stress VDS <= 5.25 V, ID <= 2 mA.
Low level at 2 mA: 2 mA x 2.0 ohm (RDS(on) max, VGS 2.5 V) = 4 mV, << 0.4 V.
R1 220R: gate charge current <= 3.3/220 = 15 mA peak; RC < 25 ns even at 100 pF, vs 20 us half-period.
Driven gate = VOH x 10k/10.22k = 3.23 V at 3.3 V (2.94 V at a 3.0 V rail); above the 2.5 V RDS(on) row.
R2 10k: GPIO Hi-Z/reset/unpowered -> gate = IGSS x 10k (<= 0.1 V at 10 uA, assumed); GPIO6: no pull at reset.
NOT rated for FAN_PWM_RAW landed on FAN_24V; verify lead identity before landing (BRINGUP step 2).
No RC filter and no fixture pull-up on FAN_PWM_RAW. JP2 open = line fully released for open-line tests.

3. Tach input / conditioning



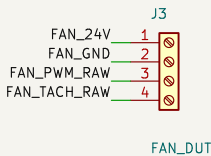
Sensor, Sanyo spec 9D0001H202: open collector.
VCE <= 27.6 V, IC <= 5 mA, VCE(sat) <= 0.8 V.
Spec suggests 2 pulses/rev; configured PPR stays 0 (unknown) until an optical RPM check on the specimen.
Default JP3 OPEN, JP4 OPEN: raw tach sees only TP8.
JP3: R3 10k to 3V3 -> 0.33 mA sink (IC max 5 mA).
JP4: R4 22k, D1 clamp, R5 470k. High -3.09 V (VIH 2.475 V); low <= 0.76 V at VCE(sat) 0.8 V (VIL 0.825 V).
Back-power if the tach lead is faulted to 26.4 V:
R4/D1 <= 1.2 mA, R3 (JP3) <= 2.7 mA into 3V3.
RULE: controller 3V3 ON before JP3/JP4 close; keep it on until fan power is off or JP3/JP4 are open.
C2 DNP (no glitch filter). Internal pull-up stays OFF.

4. 24 V fan supply and ground topology



Ground topology: single star point at J2.2 through NT1/NT2.
FAN_GND: fan supply return only. GND: Q1 source, tach clamp/pulldowns, J1.4, scope ground.
Fan motor current never flows in GND; the <= 2 mA PWM and tach currents return through the star.
Controller USB ground meets the PSU negative ONLY here: use a floating PSU output.
F1: 5x20 holder; rating TBD until startup/locked-rotor current is measured. The bench PSU current limit is the first protection during characterization (holder may carry a labelled OR link).
JP5 L_MEAS: remove link to insert an ammeter. D2/R6 (~2.2 mA) sit upstream of JP5.
No reverse-polarity element: confirm polarity at TP12 before landing the fan. C1 DNP.

5. Fan connector



J3 positions are fixture functions, not the fan's physical wire order.
Land each lead only after it is identified on the specimen by photo + continuity + manufacturer documentation.
Catalog convention / not yet verified on specimen:
red +, black -, brown PWM, yellow sensor.

6. Test points / configuration jumpers

Jumpers (Rev 0 shipping state)
JP1 GATE_EN OPEN MCU_PWM -> R1/Q1 gate
JP2 PWM_LINE_EN OPEN Q1 drain -> FAN_PWM_RAW
JP3 TACH_PU_EN OPEN R3 3V3 pull-up on raw tach
JP4 TACH_GPIO_EN OPEN raw tach -> R4/D1/R5 -> GPIO
JP5 L_MEAS FITTED 24V_FUSED -> FAN_24V

Test points
TP12 BENCH_24V TP13 FAN_24V TP14 FAN_GND
TP1, TP10 3V3 TP2, TP7, TP11 GND
TP3 PWM_IN (signal-generator injection with JP1 open)
TP4 PWM_GATE TP5 PWM_DRAIN TP6 FAN_PWM_RAW
TP8 FAN_TACH_RAW TP9 TACH_GPIO

25 kHz is the manufacturer-specified control frequency for 9GA0424P3J001; bench characterization still verifies specimen behaviour. Configured PPR = 0 (unknown).
Not defined here: JumpJet GPIOs, thresholds, startup duty, stall limits, fan-safety policy.

Specimen lead ID and configured PPR: UNVERIFIED
Rev 0: JP1-JP4 OPEN, JP5 fitted
DUT: San Ace 9GA0424P3J001 (24 V, 4-wire), drawing Rev C
BENCH characterization hardware. NOT JumpJet production.

thetechbenders / DragonBench

Sheet: /
File: fan-characterization-fixture.kicad_sch

Title: DragonBench fan-characterization fixture (BENCH ONLY)

Size: A3 Date: 2026-09-28

Rev: 0-draft

KiCad E.D.A. 9.0.9

Id: 1/1